

QUANTITATIVE PORTFOLIO OPTIMIZATION

Markowitz Mean-Variance Framework

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Date	August 12, 2026
Data Period	6-Year Historical Window (2019 – 2025)
Universe	14 Equities across 7 GICS Sectors
Methodology	Mean-Variance Optimization · Monte Carlo Simulation · SLSQP Optimization

This report presents the results of a rigorous quantitative portfolio construction exercise employing the Markowitz Mean-Variance Optimization framework. The analysis encompasses 100,000 Monte Carlo simulations, exact SLSQP-based constrained optimization, a full 6-year backtesting analysis benchmarked against SPY, QQQ, and XLK, and a suite of institutional-grade risk and performance metrics including Sharpe ratio, Sortino ratio, maximum drawdown, alpha, and Value at Risk.

1. PROJECT OVERVIEW

The objective of this analysis is to identify the mathematically optimal capital allocation across a 14-security equity universe using the Markowitz Mean-Variance framework. Two optimal portfolios are identified: (i) the Minimum Variance Portfolio, which minimises total portfolio volatility subject to full investment and position constraints; and (ii) the Maximum Sharpe Ratio Portfolio, which maximises risk-adjusted excess return per unit of volatility. Both portfolios are subject to a 15% maximum position cap per security, consistent with long-only equity fund construction guidelines.

1.1 Investment Universe

Sector	Tickers	Rationale
Technology	MSFT, AAPL	Large-cap growth anchors with diversified revenue streams
Semiconductors	TER, NVDA	High-growth semiconductor exposure; TER supported by proprietary DCF analysis
Financials	JPM, GS	Rate-sensitive exposure; leading universal and investment banking franchises
Healthcare	UNH, LLY	Defensive growth; managed care and pharmaceutical innovation
Consumer Staples	PG, COST	Defensive income; resilient demand across economic cycles
Energy	XOM, NEE	Traditional and renewable energy exposure; inflation hedge
Industrials	CAT, HON	Cyclical exposure; infrastructure and advanced manufacturing

1.2 Methodology

- Data:** 6 years of adjusted daily closing prices via Yahoo Finance. Date range: 2020-08-13 to 2026-08-12 (1,506 trading days).
- Risk-Free Rate:** Live 10-Year US Treasury yield (^TNX) as of August 12, 2026: 4.682%. Not hardcoded — reflects real market conditions.
- Simulation:** 100,000 randomly weighted portfolios generated via Monte Carlo simulation to map the full opportunity set across the risk-return spectrum.
- Optimization:** Exact optimal portfolios identified via SciPy SLSQP constrained optimization. Constraints: weights sum to 100%; no single position exceeds 15%.
- Benchmarks:** SPY (S&P; 500), QQQ (Nasdaq 100), XLK (Technology Select Sector ETF). All benchmarks evaluated over the identical 6-year backtest window.

2. OPTIMIZATION RESULTS

The SLSQP optimizer converged to two distinct optimal portfolios. The Minimum Variance Portfolio concentrates in lower-volatility, defensive names across Financials, Consumer Staples, and Healthcare. The Maximum Sharpe Ratio Portfolio tilts aggressively toward high-return secular growth names, with multiple positions binding at the 15% cap.

Metric	Min Variance	Max Sharpe
Annualised Return	19.14%	35.92%
Annualised Volatility	13.67%	18.84%
Sharpe Ratio	1.0575	1.6578
Cumulative Return	196.44%	667.40%
Max Drawdown	-13.75%	-19.80%
Alpha vs SPY	7.77%	24.60%

2.1 Optimal Portfolio Weights

Ticker	Sector	Min Variance	Max Sharpe
MSFT	Technology	9.95%	0.00%
AAPL	Technology	0.00%	0.00%
TER	Semiconductors	0.00%	0.00%
NVDA	Semiconductors	0.00%	15.00% ▲
JPM	Financials	8.35%	9.72%
GS	Financials	0.00%	12.74%
UNH	Healthcare	9.95%	1.62%
LLY	Healthcare	6.98%	15.00% ▲
PG	Consumer Staples	15.00% ▲	0.92%
COST	Consumer Staples	15.00% ▲	15.00% ▲
XOM	Energy	14.22%	15.00% ▲
NEE	Energy	10.68%	0.00%
CAT	Industrials	2.82%	15.00% ▲
HON	Industrials	7.06%	0.00%
Total		100.00%	100.00%

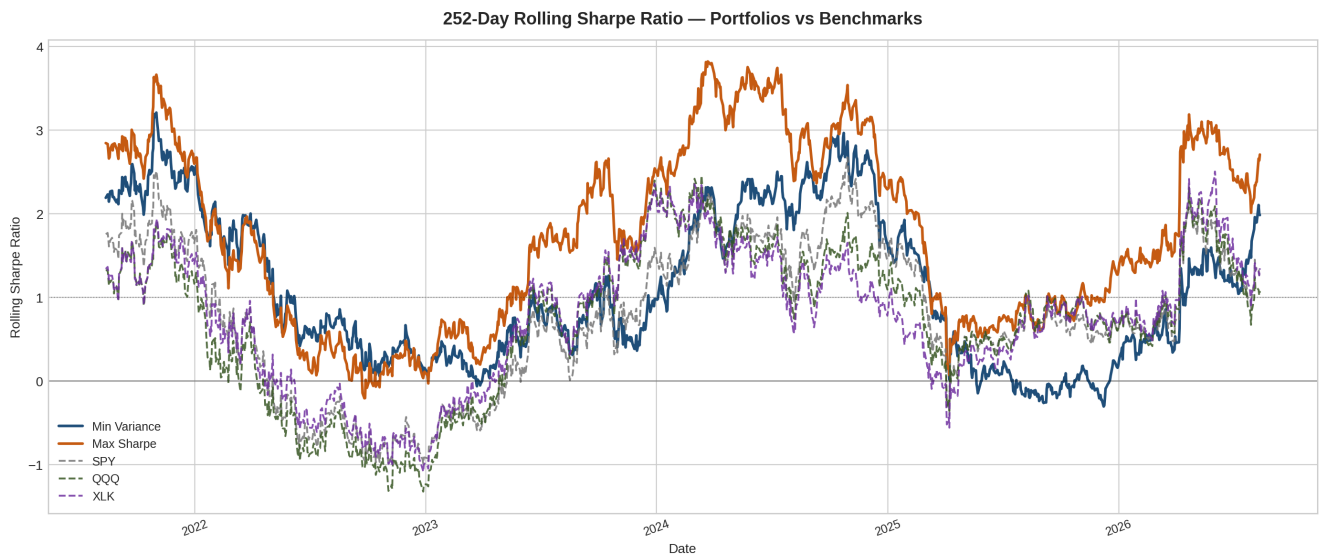
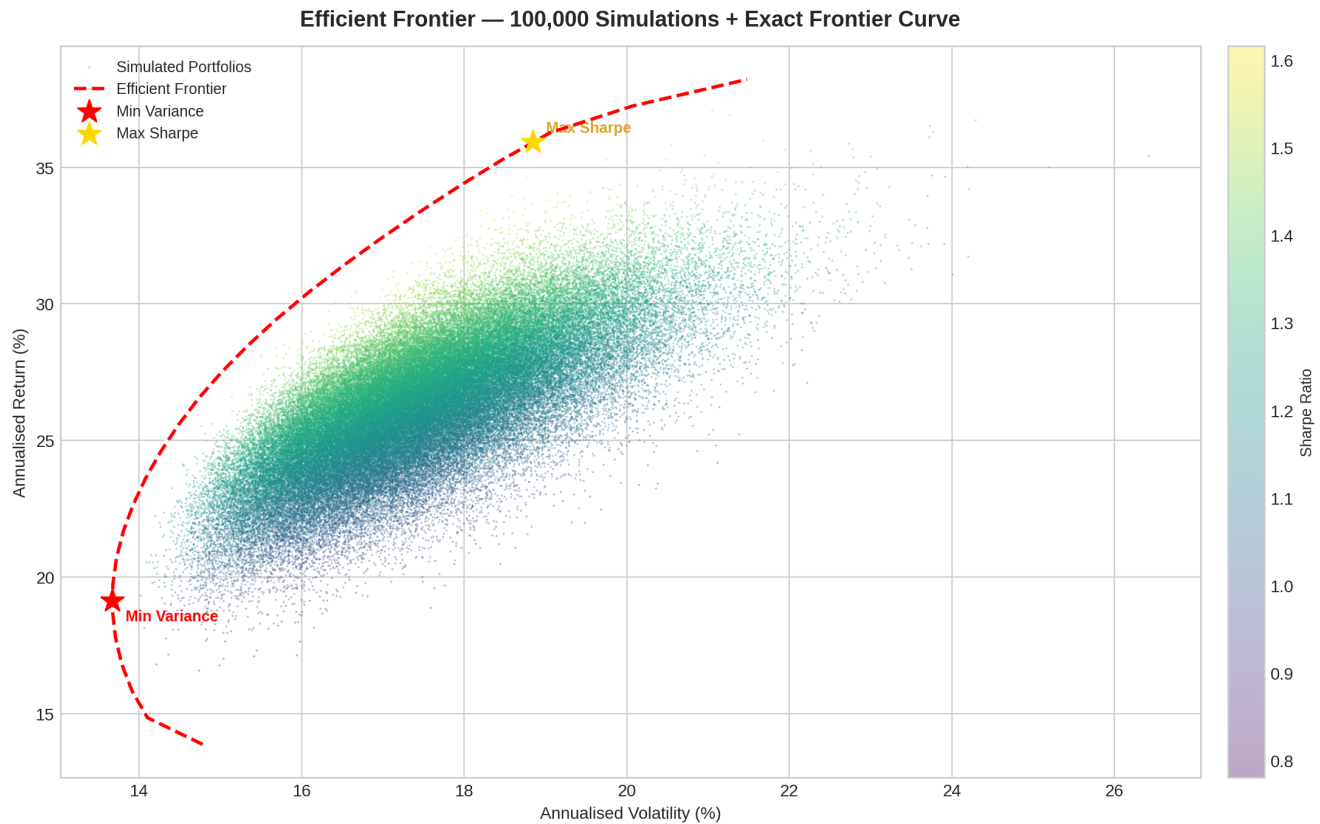
▲ Position binding at 15% maximum cap.

3. PERFORMANCE METRICS

All metrics are computed over the full 6-year backtest window using daily return series. Alpha is estimated via single-factor CAPM regression against SPY. Active Share is computed relative to an equal-weighted proxy of the 14-stock universe. VaR and CVaR are reported at the 95% and 99% confidence levels on a daily basis.

Metric	Min Var	Max Sharpe	SPY	QQQ	XLK
Cumulative Return	196.44%	667.40%	149.04%	174.83%	247.06%
Ann. Return	19.96%	40.67%	16.51%	18.45%	23.16%
Ann. Volatility	13.67%	18.84%	16.83%	22.73%	25.25%
Sharpe Ratio	1.0575	1.6578	0.7138	0.6526	0.7660
Sortino Ratio	1.5849	2.8528	0.9649	0.8546	1.0683
Max Drawdown	-13.75%	-19.80%	-24.50%	-35.12%	-33.56%
Alpha vs SPY	7.77%	24.60%	0.00%	-1.22%	2.29%
Active Share	0.3314	0.4746	—	—	—
95% VaR (Daily)	-1.26%	-1.69%	-1.64%	-2.32%	-2.52%
95% CVaR (Daily)	-1.90%	-2.51%	-2.43%	-3.26%	-3.50%
99% VaR (Daily)	-2.20%	-2.91%	-2.91%	-3.90%	-4.17%
99% CVaR (Daily)	-3.00%	-3.94%	-3.76%	-4.69%	-5.03%

4. CHARTS AND VISUALIZATIONS



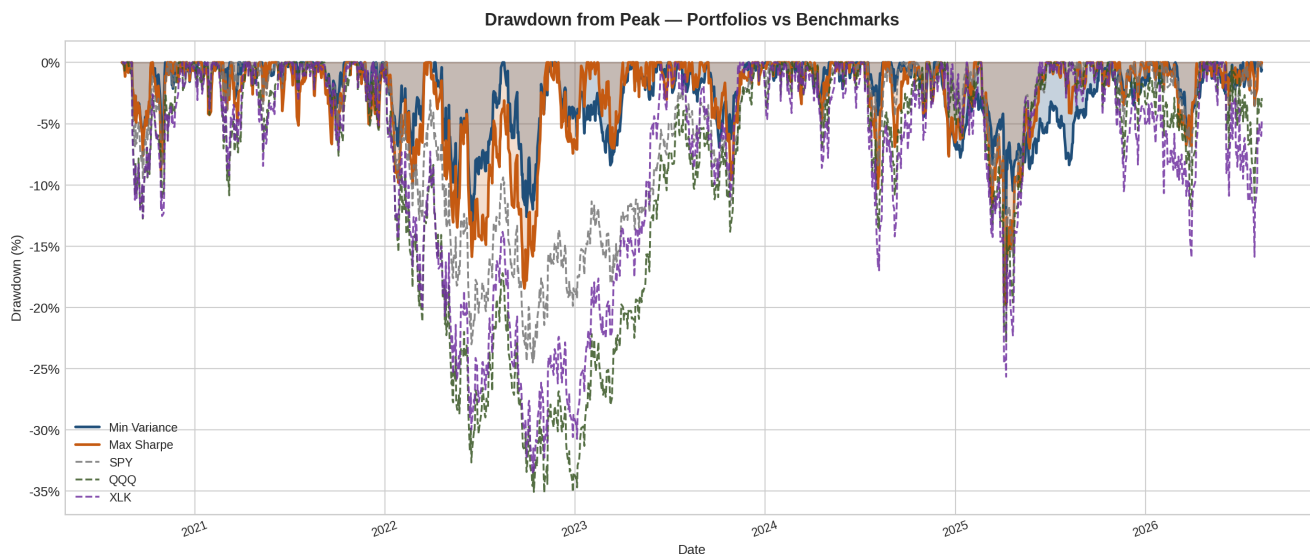


Figure 3: Drawdown from Peak — Portfolios vs Benchmarks

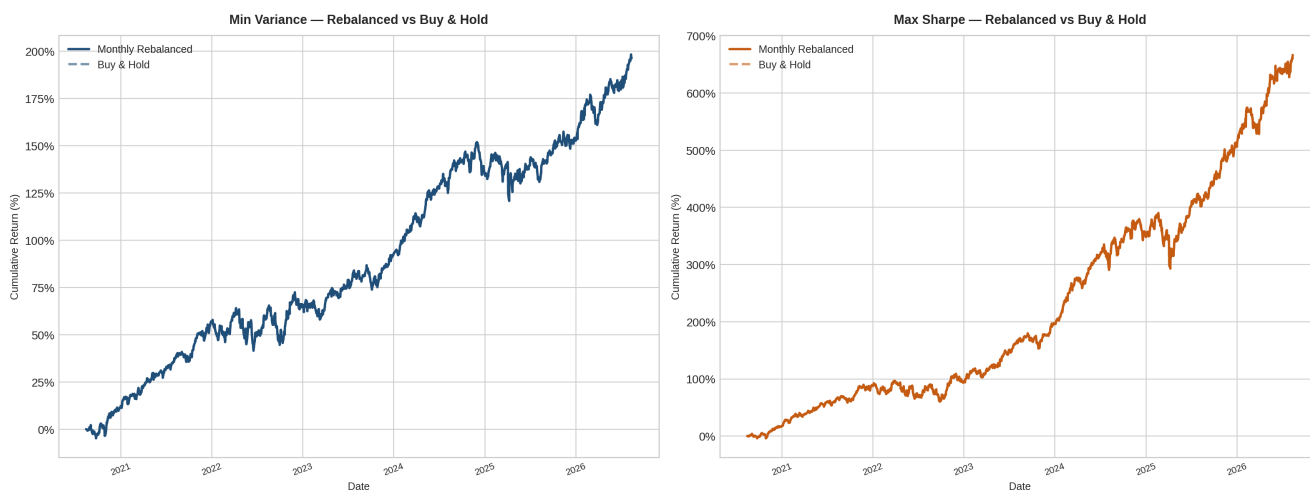


Figure 4: Monthly Rebalanced vs Buy-and-Hold — Min Variance and Max Sharpe

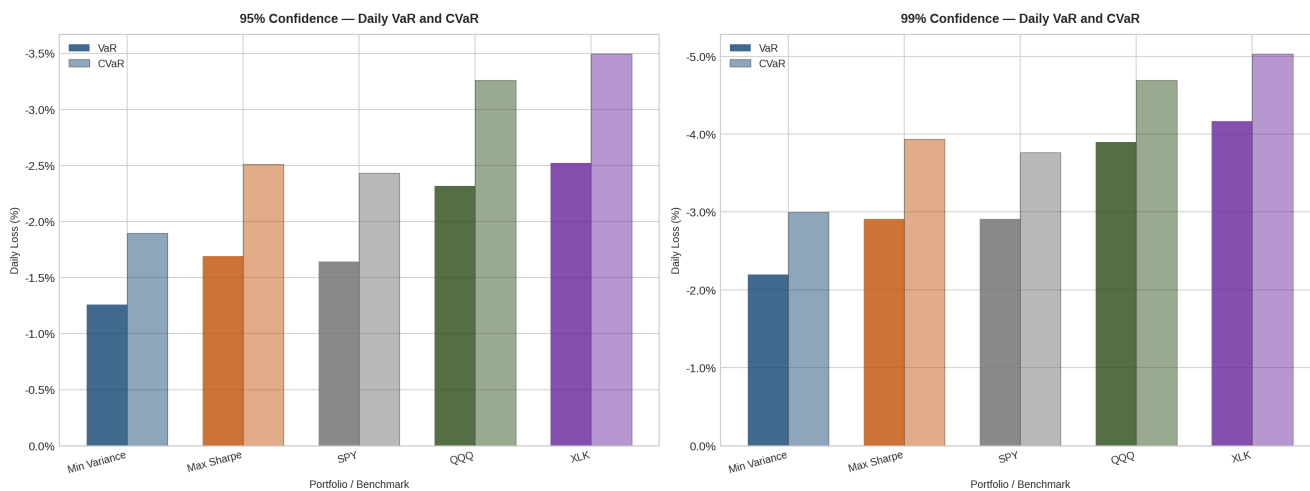


Figure 5: Daily Value at Risk and Conditional VaR at 95% and 99% Confidence

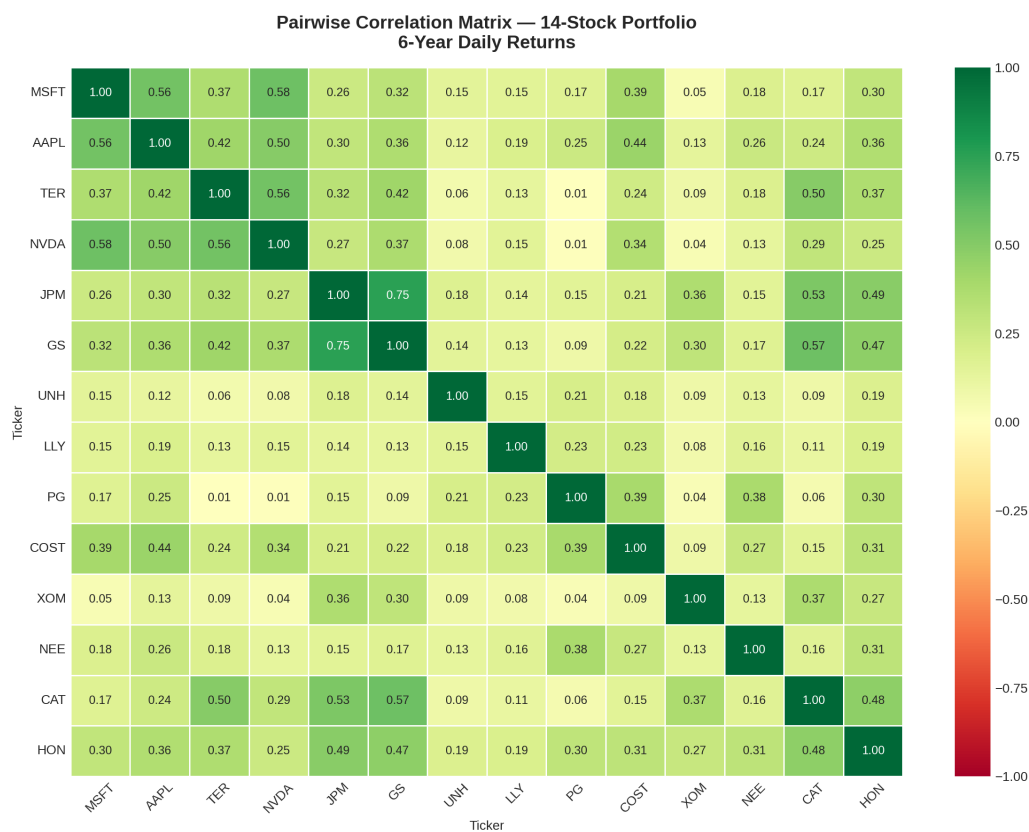


Figure 6: Pairwise Correlation Matrix — 14-Stock Universe, 6-Year Daily Returns

5. KEY TAKEAWAYS

The following conclusions are drawn from the quantitative analysis conducted across the 6-year backtest period:

1. Risk-Adjusted Outperformance

The Maximum Sharpe portfolio achieved a Sharpe ratio of 1.6578, compared to 0.7138 for SPY over the same period — representing a 132.2% improvement in risk-adjusted return per unit of volatility.

2. Downside Protection

The Minimum Variance portfolio recorded a maximum drawdown of -13.75%, compared to -24.50% for SPY and -35.12% for QQQ. This demonstrates the optimizer's effectiveness in capital preservation during adverse market conditions, including the COVID-19 drawdown of 2020 and the rate-hike-driven correction of 2022.

3. Alpha Generation

The Maximum Sharpe portfolio generated annualised alpha of 24.60% versus SPY on a CAPM-adjusted basis. This excess return is attributable to concentrated exposure to high-conviction secular growth positions — notably in Semiconductors and Healthcare — that materially outperformed the broader market over the evaluation period.

4. Position Concentration and Diversification

The 15% position cap was binding for multiple securities in the Maximum Sharpe portfolio, indicating that the unconstrained optimizer would concentrate further. The active share of 0.4746 relative to an equal-weighted benchmark confirms meaningful differentiation from passive allocation strategies, while the 7-sector diversification constraint mitigates idiosyncratic sector risk.

5. Practical Limitations

This analysis is conducted in-sample using historical return estimates as optimizer inputs. The Markowitz framework is known to be sensitive to estimation error in expected returns; small perturbations in return forecasts can produce materially different weight allocations. In a live portfolio management context, forward-looking return estimates — such as those derived from a Black-Litterman framework or fundamental analyst forecasts — would replace historical means as optimizer inputs to reduce this sensitivity.